2016 EXAMINATIONS

Lancaster University

Part I (First Year)

SCHOOL OF COMPUTING AND COMMUNICATIONS

SCC.110 Software Development (1 hour & 30 minutes)

- **Answer any THREE out of the four questions.**
- **Use a separate answer book for each question.**

Question 1

You have been employed to create software written in C for the new National Address Lottery run by Spamalot. The name and address of every person on the electoral register will be loaded into your software from a CSV (comma separated value) file and then the program should randomly select a winning name and address each week. The previous programmer has been sacked because he won the lottery in the first week. You need to take over his code and fix the bugs that he has left behind.

A small test sample input file called "addresses.csv" is shown below:

```
Miss Scarlet,23 Red Lane,Leicester
Professor Plum,Purple Cottage,Plumpton
Mrs Peacock,1 Blue Street,Bloomington
Reverend Green,The Vicarage,Churchtown
Colonel Mustard,59 Letsby Avenue,Yellow Garden City
Mrs White,7 Cloud Grove,Isle of Wight
```

1a. There are eight errors in the code overleaf that are stopping the software from compiling or running correctly. List any five of these errors and state the line numbers on which they occur.

[5 marks]

1b. Describe the consequences of each of the five errors that you identified in part 1a and state whether they cause problems during compilation or execution of the code. Finally, suggest a correction for each error.

[10 marks]

Question 1 continues on next page…

Question 1 continued

```
1     #include <stdio.h>
2     include <stdlib.h>
3     #include <string.h>
4
5     #define NUMPEOPLE 100;
6     #define LINELENGTH 100
7
8     struct person {
9     char name[20];
10    char street[20];
11    char *city;
12    };
13
14    int main (
15    {
16    struct person *people;
17    int i, j, p, length;
18    FILE *myFileHandle;
19
20      people = (struct person *)malloc(sizeof(struct person)*NUMPEOPLE);
21
22      myFileHandle = fopen("addresses.csv", "r");
23
24      if (myFileHandle == NULL) {
25        char lineOfText[LINELENGTH];
26
27        while (fgets(lineOfText, LINELENGTH, myFileHandle) != NULL) {
28          printf("Line read is: %s", lineOfText);
29          length = stringlength(lineOfText);
30          i = 0;
31          j = 0;
32          // name
33          while (lineOfText[i] != ',') {
34            people[p].name[j] = lineOfText[i];
35            i++;
36            j++;
37          }
38          people[p].name[j] = '\0';
39          // street
40          i++;
41          j = 0;
42          while (lineOfText[i] != ',') {
43            people[p].street[j] = lineOfText[i];
44            i++;
45            j++;
46          }
47          people[p].street[j] = '\0';
48          // city
49          i++;
50          j = 0;
51          while (lineOfText[i] != '\n') {
52            people[p].city[j] = lineOfText[i];
53            i++;
54            j++;
55          }
56          people[p].city[p] = '\0';
57          p++;
58        }
59        fclose(myFileHandle);
60      }
61    }
```

Figure for Q 1.a.

Question 1 continues on next page…

Question 1 continued

1c. In order to check that your corrected code has loaded in the test file correctly, write in C, some code to print out the contents of the data in exactly this format with a new line for each person:

```
Number of people read: 6
Person 0: name = 'Miss Scarlet' street = '23 Red Lane' city = 'Leicester'
Person 1: name = 'Professor Plum' street = 'Purple Cottage' city =
'Plumpton'
Person 2: name = 'Mrs Peacock' street = '1 Blue Street' city =
'Bloomington'
Person 3: name = 'Reverend Green' street = 'The Vicarage' city =
'Churchtown'
Person 4: name = 'Colonel Mustard' street = '59 Letsby Avenue' city =
'Yellow Garden City'
Person 5: name = 'Mrs White' street = '7 Cloud Grove' city = 'Isle of
Wight'
```

Your new code should be able to be inserted in the corrected program after the fclose statement on line 59.

[5 marks]

1d. The randomisation code from the original programmer has mysteriously been deleted. Using pseudocode only but with your knowledge of the necessary steps in C, describe a step-by-step algorithm that randomly selects a winner from the list of people who entered the lottery. The program should not always return the same winner when run.

[5 marks]

[Total 25 marks]

Question 2

Imagine you have been asked to write software in Java to assist in tracking the status of endangered animal species. The software is required to hold information about species and their conservation status. Based on this data, the software is also required to provide simple summaries and scientific predictions.

2.a Write a Java class to represent an animal species. This class should contain instance variables to hold the following pieces of information. Be careful to choose the best types to hold this information, and ensure your class is properly encapsulated.

- The name of the animal species.
- The current global population (total number of animals in existence).
- The population annual growth rate (e.g. 0.5 would mean the population is halving every year).

[4 marks]

2.b Write an appropriate constructor for your class.

[4 marks]

2.c Show how you would create instances of your class to hold the following information.

Name	**Population**	**Growth Rate**
Cheetah	9,000	0.5
Sumatran Tiger	400	0.8
Blue Whale	50,000	0.9
White Rhino	3	1.1

[2 marks]

Question 2 continues on next page...

Question 2 continued

2.d The World Wildlife Fund (WWF) maintain a range of classifications to indicate just how threatened a species is with extinction, including the following. In all cases, the LOWEST row in the table that matches a given species is the one that applies:

Classification	Definition
LC	Least Concern. Species has a growing population of at least 30,000.
VU	Vulnerable. A species has a declining population (a growth rate that is less than one)
EN	Endangered. A population is below 30,000.
CR	Critically Endangered. A population is below 500.
EX	Extinct. No remaining animals are known to exist.

Write a method for your class that will calculate the classification of that animal species, and return it from the method. Choose an appropriate name and return type for your method.

[6 marks]

2.e From the data provided, you have been asked to write code to predict how long (in years) a species will survive before becoming extinct. n.b. The Growth rate for a species defines how many more (or less) animals there will be each year. Therefore a growth rate of 2.0 means that a species population is doubling in size every year. A growth rate of 0.5 means the species populations is halving every year.

Write a method for your class that can predict how many years it will be before that species will become extinct. Also, write a Javadoc comment for your method, and be careful to clearly define in there any assumptions made by your code.

[4 marks]

2.f Show how the object instances you created in part c) could be added into an array. Write code that displays to the screen the conservation status and predicted time to extinction for each item in the array.

[5 marks]

[Total 25 Marks]

Question 3

3.a A software process is a structured set of activities required to develop a software system. List and briefly explain four main activities of a software process.

[4 marks]

3.b Incremental development is an example of a software process model. List and briefly explain two problems of the incremental development model.

[2 marks]

3.c List three types of testing you may need to perform in addition to the functional testing.

[3 marks]

3.d Consider a program, which should take a number as input and output its square root. Provide four equivalent partitions and their test cases to test the program.

[4 marks]

3.e Consider the method provided in the text box below:

```
void encrypt(int key)
{
        String enc = "";

        for (int i=0; i<message.length(); i++)
        {
                char c = message.charAt(i);
                c = (char) (c + key);

                if (c > 'z') c = (char) (c - 26);

                enc = enc + c;
        }
        message = enc;
}
```

i. Draw a flow graph for the code. Label the edges of the flow graph with T (true) and F (false) if appropriate. Copy the code to your answer-book and mark within the method to indicate which source code parts belong to which node of your flow graph.

[8 marks]

Question 3 continues on next page...

Question 3 continued.

ii. Calculate the cyclomatic complexity of the graph.

[1 mark]

iii. Describe all independent paths using the node numbers, {T} and {F}.
(e.g. Path 1: 1, 2{F}, 5 / Path 2: 1, 2{T}, 6 / ...)

[3 marks]

[Total 25 Marks]

Question 4

4.a Consider the following Java program:

```
1   public class Something
2   {
3       public int doSomething(String s1, String s2)
4       {
5           int c;
6           boolean m;
7
8           c = 0;
9
10          for (int i=0; i < s1.length(); i++)
11          {
12              m = false;
13              for (int j=0; j < s2.length(); j++)
14              {
15                  if (s1.charAt(i) == s2.charAt(j) && m == false)
16                  {
17                      System.out.println(s1.charAt(i));
18                      c++;
19                      m = true;
20                  }
21              }
22          }
23
24          return c;
25      }
26
27      public static void main (String[] args)
28      {
29          Something s1 = new Something();
30
31          String a = "lancaster";
32          String b = "science";
33
34          int r = s1.doSomething(a,b);
35          System.out.println(r);
36      }
38  }
```

Question 4 continues on next page...

Question 4 continued

i. Describe as precisely as you can the meaning of the code on the following lines:

- Line 1
- Line 3
- Line 6
- Line 24
- Line 29
- Line 34

[6 marks]

ii. Analyse the above program. Describe as precisely and concisely as you can what the program does. Write down what the program would print to the screen.

[7 marks]

4.b Professional software developers apply a set of style guidelines when writing their code in order to make their code easier to maintain, read and share with others. The following code snippet has been written by a poor programmer:

```
int main ()
{
     int i=0, e, n, s=0;float a;char c;
  scanf ("%d", &e);
  while (i < e) {scanf ("%d", &n);
         s=s+n;
i++;}
     a=s/e;
  printf ("e = %d, s = %d, a = %f.\n", e, s, a);      }
```

i. Find and list five stylistic faults in this code.

[5 marks]

ii. Rewrite the code according to the SCC110 style guidelines.

[5 marks]

iii. There are three variable types in this code. Briefly describe these three types of variables and the differences between them.

[2 marks]

[Total 25 Marks]

END OF PAPER